## Achievements

- Finished **Top 15** among **200+ UG/PG students** in **QRT QuantQuest**, **IIT Bombay** (sole PG team) *(2024)*
- Secured **2nd place** in **F.I.R.E.** (Flame Imaging Robot Extinguisher) technical event at **Tirutsava** *(2020)*
- Secured **2nd place** in **ArduiKnow Challenge**, a competitive microcontroller event at **Tirutsava** *(2020)*
- Secured **3rd place** in **Terrace Farming Robot** design contest among **23 IITs** at **Inter IIT Tech Meet** *(2019)*
- Won **1st place** in **The Line Following Robot** algorithm design event by **TechManiacs**, **IIT Tirupati** *(2019)*

## Professional Experience

**Software Developer (Backend)** | *Paytm* *(Jan '23 - Dec '23)*
*FSM & BINARY Teams, Paytm, India*

- Architected hierarchical, location-based **feature-flagging** backend, accelerating product feedback loops by **35-40%**
- Developed dynamic **territory access controls**, incentivizing agent productivity to drive a **19% increase** in resolutions
- Engineered **API orchestration** to cross-reference KYC databases, reducing vendor onboarding friction by **32%**
- Designed load-balanced **round-robin** automated ticketing algorithm, saving team leads **15+ hours/week** of overhead
- Automated manual finance workflows by **90%** using efficient **1-SAP logic** to process **15M revenue records** monthly

**Jr. Data Scientist** | *Holmusk* *(Jun '22 - Aug '22)*

- Architected predictive patient outcome models within the **NeuroBlu** webapp using **XGBoost and LSTM** architectures to model structured clinical and longitudinal patient data, improving prediction accuracy for healthcare analytics
- Optimized dashboard backend using **Python**, reducing query latency from **20s to 2s** (**10x speedup**)

**AI Research Intern** | *Motilal Oswal Financial Services* | *Internship* *(May '26 - Aug '26)*
*Align Before You Quantize: Time Series Tokenization*

- Engineered **HALO** framework for **Leech lattice** time-series tokenization via geometric alignment for **AAAI**
- Reduced reconstruction MSE by **11.9%**(IN) & **2.8%**(US) via phased training, boosting codebook usage **3.89x**
- Enhanced downstream forecasting via discrete representations, achieving SOTA **IC** & cutting **MASE** by **7.5%**

## Research Experience

**PARAM 1 Indic-Scale Foundation Model** | *BharatGen* *(Jan '25 - May '25)*
*Guide: Prof. Ganesh Ramakrishnan,* ***IIT Bombay***

- Developed **PARAM-1**, a **2.9B parameter** English-Hindi foundation model trained from scratch on **5T tokens**
- Processed **12TB text** via **NVIDIA NeMo**, deploying fastText classifiers & GPU-accelerated fuzzy deduplication
- Automated web scraping for archive.org via custom scripts, executing strict code/math filtration & **PII removal**

## Key Projects

**Implied Volatility Surface Forecasting** *(Aug '25 - Aug '26)*
*Master's Thesis | IITB Citadel Research Lab Project |(Submitted to 7th Int. Conf. on AI in Finance)*

- Spearheaded research on no-arbitrage implied volatility surface forecasting using GANs and Diffusion Models, evaluating **AF-TransVol** and **AF-INO** architectures across **3,417 trading days** of NIFTY50 data
- Engineered **44 causal scalars** & **3 soft regimes** using scale-invariant log-IV transforms & $(\rho, z)$ warping
- Enforced butterfly convexity & calendar monotonicity, achieving SOTA **5.30% MAPE** and $R^2 = \mathbf{0.910}$
- Slashed static arbitrage violations by $>$**50%** using stress-conditioned **AF-TransVol-Stress** extensions

**Interpreting Emotion & Persistence Dynamics in LLMs** | *Course Project* *(Jul '25 - Nov '25)*
*Guide: Dr. Pushpak Bhattacharyya | CS 772 : Deep Learning for Natural Language Processing*

- Traced emotion across four **Qwen2.5** LLMs via logistic probes, revealing separability peaks in mid-to-late layers
- Deployed Streamlit app mapping causal tokens via activation patching, token ablation & layer-wise noise injection
- Curated **7,000-sample** dataset and formulated decay protocol yielding **100+** token emotional half-life
- Fine-tuned transliteration via **LoRA**, hitting SOTA **15.0% CER** & **80.0% Word Accuracy** on noisy data

**Multi-Task Image Restoration with LoRA-DINOv3** | *Course Project* *(Jul '25 - Nov '25)*

*Guide: Dr. Biplab Banerjee | GNR 650 : Advance Deep Learning for Image Analysis*

- Integrated frozen **DINOv3 ViT** backbone with **LoRA** & dual **PixelShuffle** decoders for joint upscaling
- Applied global residual learning with **VGG perceptual loss**, eliminating checkerboard artifacts effectively
- Executed curriculum unfreezing at epoch 10, achieving **31.22 dB PSNR** & **0.8424 SSIM** on DIV2K SR
- Preserved scene content across enhancements, attaining **16.80 dB PSNR** & **0.7635 SSIM** on MIT-Adobe-5K

**Dense-Residual Transformer for Image Super-Resolution** | *Course Project* *(Jan '25 - May '25)*

*Guide: Dr. Biplab Banerjee | GNR 638 : Machine Learning for Remote Sensing II*

- Architected **DRCT** networks utilizing **Swin-Transformers** & **0.2 residual scaling** to resolve deep bottlenecks
- Designed 3-phase progressive loss blending **L1, MSE, SSIM** across milestones to optimize edge quality
- Accelerated training via PyTorch **DDP** on 90/10 data splits, maximizing joint **40×SSIM + PSNR** metric
- Stabilized forward spatial information flow, preventing degradation & extracting sharp high-frequency textures

## Other Projects

**Tick-Ops: Quant Data Engine Evaluation** | *Self Project* | *DuckDB, QuestDB* *(Jan '26 - May '26)*

- Benchmarked **DuckDB, QuestDB, ClickHouse** against **Pandas** across 5 tick workloads with **1.05M quotes**
- Built deterministic verification oracle enforcing strict float tolerances (**1e-12**) to ensure exact accuracy
- Demonstrated **19.5x compute speedups** via **DuckDB**, leveraging sorted row-groups to skip data partitions
- Validated order book state integrity, resolving boundary leaks using monotonic sequence IDs & timestamping

**Lock-Free MPMC Queue** | *Self Project* | *Low-Latency C++20* *(Aug '25 - Nov '25)*

- Architected high-throughput **MPMC queue** for UDP feeds, strictly isolating network I/O from payload parsing
- Implemented **C++20 lock-free ring buffer** utilizing **CAS atomics** & memory barriers to block race conditions
- Eliminated OS-level thread blocking via per-slot sequence numbers, ensuring highly concurrent memory access
- Profiled latency via **Python**, slashing median wait times by **56%** and p99 tail latency to **25µs**

**Derivatives Pricing & Volatility Research Platform** | *Self Project* | *JAX, NumPy* *(Jan '26 - May '26)*

- Decoupled financial logic to build a model-agnostic pricing platform via **Python** & **JAX** for JIT compilation
- Formulated discrete cash-dividend treatments via piecewise lognormal models to accurately price equity options
- Optimized numerical execution stacks, benchmarking binomial valuations at ∼**910µs** & local-vol repricing at **252ms**
- Enforced static no-arbitrage constraints, generating theoretical quotes within live spreads **34%** of the time

**Dental Crown Marketplace MVP** | *Self Project* | *React Native, Next.js, PostgreSQL* *(Jan '26 - Aug '26)*

- Engineered a **zero-server-overhead direct file-streaming pipeline** to anonymous storage endpoints, handling **100MB+ binary CAD scans** via direct client-to-host multi-part uploads
- Mitigated concurrent job acceptance race conditions utilizing **Prisma ORM** row-level database locking logic
- Dispatched real-time job broadcasts via **Firebase Cloud Messaging**, replacing WebSockets to slash overhead
- Secured manufacturing accountability by tracking every state transition within an **immutable SQL audit ledger**

**Lock-Free Count-Min Sketch** | *Self Project* | *C++, Atomics, BusTub* *(Aug '26)*

- Engineered a lock-free frequency sketch using **C++ atomics**, achieving a **2.16x speedup**
- Optimized throughput via **memory_order_relaxed**, eliminating contention on isolated hash counters
- Resolved move-semantics hazards by regenerating hash closures, ensuring strict ASAN/LSAN safety
- Delivered deterministic TopK queries, attaining **100% correctness** in 13 multi-threaded tests

## Positions of Responsibility

**Teaching Assistant**: *DS203: Programming for Data Science & EPGD, IIT Bombay* *Jan '26 - Aug '26*

- Automated evaluation scripts & allocated cloud credits, streamlining grading for **110 students**
- Designed evaluation prompts against rubrics, managed grading metrics & reviewed notebooks

**System Administrator**: *CMInDS Department, IIT Bombay* *Jul '24 - Dec '25*

- Deployed **OSTicket**, new features to Interview Portal & Research portals to centralize digital department services
- Engineered automated backups and maintained infrastructure stability for continuous operations

**Placement Representative**: *Indian Institute of Technology Tirupati* *Jul '21 - Jun '22*

- Facilitated recruitment for **170+ students** by coordinating logistics & talks, securing vital industry partnerships

## Technical Skills

| | |
|---|---|
| **Languages** | Python, C/C++, JavaScript, Java, Bash, SQL, Markdown |
| **ML & Data Science** | PyTorch, TensorFlow, JAX, Scikit-Learn, Keras, OpenCV, NumPy, Pandas, SciPy |
| **Web & Utilities** | Django, Spring Boot, React.js, Flask, Matplotlib, Seaborn, BeautifulSoup, Selenium |
| **Tools & Platforms** | Git, Docker, AWS, Linux, MATLAB, LaTeX, Pygame, p5.js, InkScape, Procreate |

## Extracurricular Activities

- Active in **Drawing**, **Table Tennis**, and **Brand Logo Designing**